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Sugar Daddy: Quotas and the U.S. Government

In 2001 the impending closure of a large Brach's candy factory in Chicago sparked a war of words between Mayor Richard M. Daley and the powerful U.S. sugar lobby. The candy industry employs roughly 10,000 people in the greater Chicago area. However, while sugar is a major industry in the Chicago area, Illinois is also a major producer of high fructose corn syrup, a substitute sweetener that has benefited from the government's sugar program. This has plunged the mayor into a classic upstate-downstate battle with corn growers and agribusinesses in Illinois. There is much at stake, because Congress is beginning to review farm supports on sugar and a number of other agricultural commodities. Senator Richard Durbin, a downstate native, notes that it will be a difficult vote for Illinois senators because "you have important people on both sides of the issue."

Your firm has been drafted by city hall to prepare a report detailing the costs of the government's sugar program and its effect on Chicago industry. A major Senate vote on the sugar program is scheduled next week, so your project leader has set a deadline for the report to be completed within 72 hours.

How the U.S. Sugar Subsidy Program Works

Since 1981 the U.S. federal government has operated a price support program to stabilize sugar beet and sugar cane producers and processors by maintaining high sugar prices. The government is able to support prices because sugar is a deficit crop (i.e., the United States consumes more than it produces). Thus, the government is able to maintain prices by limiting imports.

The U.S. government manages sugar subsidies through a loan program. Loans are made directly to sugar processors rather than to farmers. To be eligible for a loan, processors must pay producers a government-specified minimum price for sugar beets or sugar cane. Processors pledge sugar as collateral to obtain a loan from the government. Loans must be repaid within nine months or by the end of the fiscal year; this repayment scheme ensures the program does not appear on the federal budget at the end of the fiscal year. When the loan matures, processors have two choices. They can either pay off the loan plus interest or forfeit the sugar and keep the government's money.

The United States Department of Agriculture (USDA) is not in the sugar business, and does not like to purchase sugar. It restricts imports to maintain high domestic prices for sugar so that processors have an incentive to redeem their sugar at the end of the fiscal year. Each quota year the USDA establishes a tariff rate quota (TRQ) for sugar, which is divided among the countries that export sugar to the United States. Sugar can enter the United States in excess of the TRQ, but a prohibitively high duty of 16 cents per pound is imposed on imports in excess of a country's quota. This duty is sufficiently high to make sugar imports unprofitable.

In 1998 the countries with the three largest quotas were the Dominican Republic, Brazil, and the Philippines (**Exhibit 1**). Roughly speaking, the quotas are set in proportion to the distribution of imports over 1975–1981. However, the quotas are often adjusted to serve political ends. For instance, Cuba lost its quota in 1961. The quota for Nicaragua was reduced to zero after the Sandinistas took power, but was restored in 1990 after the government changed. Nicaragua is now one of the leading sugar importers to the United States. Sugar quotas are also sometimes increased for developing countries.

During the Uruguay Round, the United States committed itself to permit minimum imports of roughly 1.25 million tons of sugar per year. Under the North American Free Trade Agreement (NAFTA), Mexico will be entitled to growing imports in future years.

The Sugar Market

Over the past 20 years U.S. consumers have typically paid twice the world market price for sugar (**Exhibit 2**). Recently, the disparity has been even higher. As of December 2001 U.S. consumers paid 22.9 cents per pound, while the world price was 9 cents per pound. Due to expanding production, world prices are likely to remain low. Global sugar production has increased 22 percent in the last six years.

The elasticity of demand for sugar has been estimated at -0.3 and the elasticity of supply has been estimated at 0.5 .¹ The U.S. General Accounting Office (GAO) estimates that the total cost of the subsidy program to consumers is \$1.9 billion per year. However, sugar growers disagree. They say such estimates are inflated because candy makers do not pass along savings in raw materials costs to consumers. The U.S. International Trade Commission concluded that abolishing the program would result in a net welfare increase of about \$986 million.

Due to the combined effects of new technologies, favorable weather conditions, and the government support program, U.S. sugar production has recently surged ahead of demand. **Exhibit 3** shows the rapid increase in sugar production throughout the 1990s. In the late 1990s production increased, on average, 6 percent per year, while demand grew less than 2 percent per year. In 2000 total U.S. production was projected at 14.4 billion pounds while U.S. consumption was 17.4 billion pounds.

In response to the continuing oversupply, the USDA has recently been forced to take steps to boost prices. It bought more than one million tons of sugar for \$435 million in 2000. This excess sugar is currently stored in a warehouse at a cost of \$1.4 million per month. In August 2000 the

¹ These elasticity estimates are taken from Vernon Roningen and Praveen Dixit, "The Potential Impacts of Liberalized Trade Policies in Industrial Market Economies," U.S. Department of Agriculture, Economic Research Service Staff Report No. AGES 89-36: August 1989; and F. M. Sherer, "The United States Sugar Program," Kennedy School of Government Case Study, Harvard University, 1992.

USDA ordered sugar growers to plow under some of their sugar beet crop. These actions were not able to drive prices above the “strike price” necessary for producers to pay back their loans. Consequently, in 2000 the USDA paid out large sums of money for sugar for the first time since 1986.

Who Benefits From the Quota System?

Proponents of sugar subsidies argue that 420,000 jobs in the U.S. economy are tied to the sugar industry. However, skeptics argue that this number is probably inflated. The U.S. International Trade Commission argues that 16,400 workers are involved directly in sugar production (raw cane sugar, beet sugar, cane sugar refining). This production is concentrated in four sugar cane-producing states and 14 sugar beet-producing states (**Exhibit 4** and **Exhibit 5**). The remaining jobs represent workers in related industries.

One industry that benefits from artificially inflated sugar prices is high fructose corn syrup (HFCS). HFCS has a similar molecular structure to sugar, and in syrup form has nearly the same sweetness. HFCS production diffused rapidly in the 1970s after technological revolutions in Japan. Corn syrup producers benefit from sugar subsidies because the high prices for sugar create incentives for food producers such as soft drink manufacturers to use corn syrup as an alternative sweetener. HFCS has consistently sold at a substantial discount to sugar. In addition to its usage in soft drinks, HFCS has been employed in fruit canning, baking, and breakfast cereals. It is the presence of downstate HFCS producers that has created a conundrum for Illinois politicians; they have been unable to decide whether to support candy producers or the corn syrup industry.

Trade Implications

The sugar program has had major implications for U.S. trade policy. Other countries point out that U.S. agricultural programs are hypocritical, and these policies are widely seen as a barrier to the negotiation of new free trade agreements. These countries argue that U.S. negotiators press other countries to open their markets to services and technology industries while simultaneously maintaining a protectionist stance on commodities. Even former U.S. Trade Representative Charlene Barshefsky has said the United States will have to address some of its “Achilles’ heels, such as textiles and sugar” if it wishes to inaugurate a new round of world trade talks.

The sugar program was a major point of contention during discussions for the proposed Free Trade Area of the Americas (FTAA) pact. Prior to the 2001 Summit of the Americas, representatives from Brazil argued it would join a FTAA only if the United States opened up its markets for agricultural and steel commodities and modified its anti-dumping laws. “If the FTAA is not sacred-cow-hunting season, then we’re not going to have the FTAA,” proclaimed Pedro de Camargo, the head of the Brazilian Rural Society, Brazil’s second-largest farm group.

Discussion Questions

What evidence would you present to lobby against U.S. sugar policy? Your advisors suggest that the following evidence should be presented in your report: (a) the effect on consumers of sugar, (b) the effect on domestic producers of sugar, and (c) the effect on foreign producers of sugar. To help generate your report, you are invited to work through the problems below.

1. Assume that the annual U.S. demand for sugar is linear, and that demand and supply for sugar were identical in 2000 and 2001. Use the information in the case to find the following information to support your analyses. For simplicity, you may work with demand and supply curves that aggregate cane and beet sugar.
 - (a) Find the demand and supply curves for raw sugar.
 - (b) Under the subsidy policy, find the U.S. farmers' production and total imports into the United States.
 - (c) Under the subsidy policy, calculate the change in welfare for U.S. consumers of sugar, domestic producers, and foreign producers. For the purposes of this question, you may assume that all demand for sugar is final demand.
2. Would sugar-exporting countries be better or worse off if the United States replaced its import quotas with tariffs sufficient to balance supply with demand at a domestic raw price of 22.9 cents per pound? How would sugar-exporting countries fare if this tariff revenue were then used as foreign aid to less developed countries (LDCs)?
3. What effect would a relaxation of sugar subsidies have on local candy producers? Sugar growers argue that candy producers would not pass along cost declines to candy consumers. Do you agree or disagree?
4. The sugar lobby commonly argues that U.S. sugar subsidies are less protectionist than those of other countries, especially Europe and Japan. Supporters argue that because of their less protectionist nature, these policies save Americans \$2 billion per year. Jack Roney, an economist with the American Sugar Association, has said growers want subsidy policies retained until other governments remove their policies as well. He said, "What we are saying is, 'Let's keep these import controls until others have dismantled theirs, and then we'll join them.' " How would you respond to these statements?
5. Why do you think it is so difficult to repeal the sugar program?



Exhibit 1: U.S. Tariff Rate Quotas, 1998

Nation	Quantity
Dominican Republic	591.6
Brazil	487.4
Philippines	453.8
Australia	279.0
Guatemala	161.4
Argentina	144.6
Peru	137.8
Panama	97.4
El Salvador	87.4
Colombia	80.6
South Africa	77.4
Nicaragua	70.6
Mexico	55.2
Swaziland	53.8
Costa Rica	50.4
Other Caribbean	109.8
Other South American	132.8
Other African	222.0
Other Asia and Pacific	160.4

Note: Quantities are in millions of pounds.

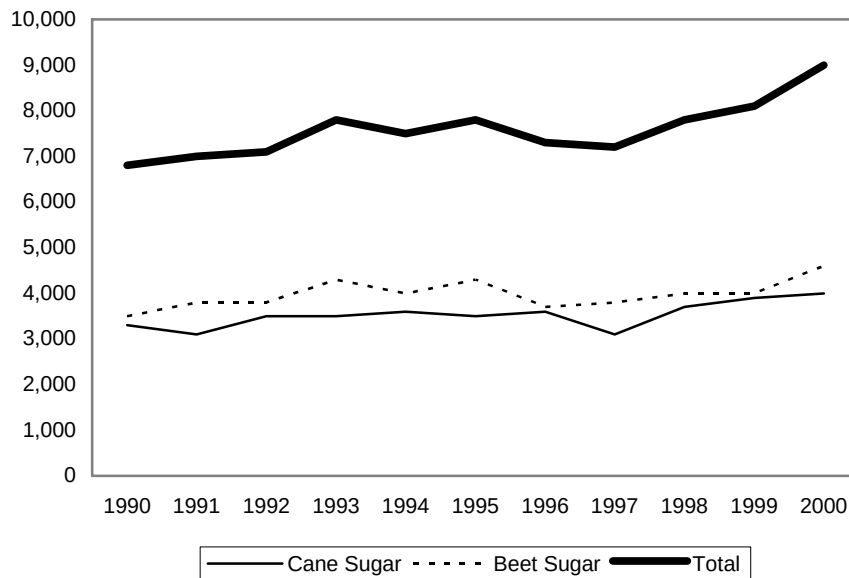
Source: Sugar and Sweetener Situation and Outlook. Economic Research Service, USDA (September 1999) and the New York Coffee, Sugar, and Cocoa Exchange.

Exhibit 2: U.S. and World Prices for Raw and Refined Sugar, 1985–1998

Year	U.S. Raw Cane Sugar	U.S. Wholesale Refined Beet Sugar	U.S. Retail Refined Sugar	World Raw Sugar	World Refined Sugar
1985	20.34	23.18	35.34	4.04	6.79
1986	20.95	23.38	35.08	6.05	8.47
1987	21.83	23.60	35.28	6.71	8.75
1988	22.12	25.44	36.60	10.17	12.01
1989	22.81	29.06	40.03	12.79	17.16
1990	23.26	29.97	42.78	12.55	17.32
1991	21.57	25.65	42.80	9.04	13.41
1992	21.31	25.44	41.53	9.09	12.39
1993	21.62	25.15	40.54	10.03	12.79
1994	22.04	25.15	39.99	12.13	15.66
1995	22.96	25.83	39.83	13.44	17.99
1996	22.40	29.20	41.79	12.24	16.64
1997	21.96	27.09	43.26	12.06	14.33
1998	22.06	26.12	42.98	9.68	11.59

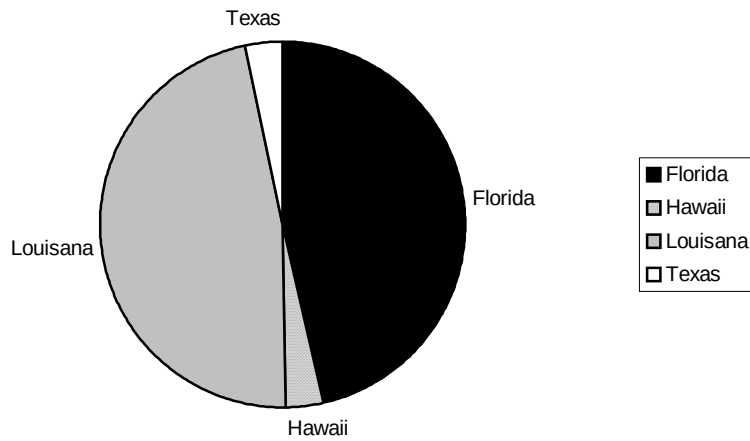
Notes: Prices in cents per pound. U.S. and world prices in nominal dollars.

Source: Sugar and Sweetener Situation and Outlook. Economic Research Service, USDA (September 1999) and the New York Coffee, Sugar, and Cocoa Exchange.

Exhibit 3: U.S. Sugar Production (in thousands of short tons)

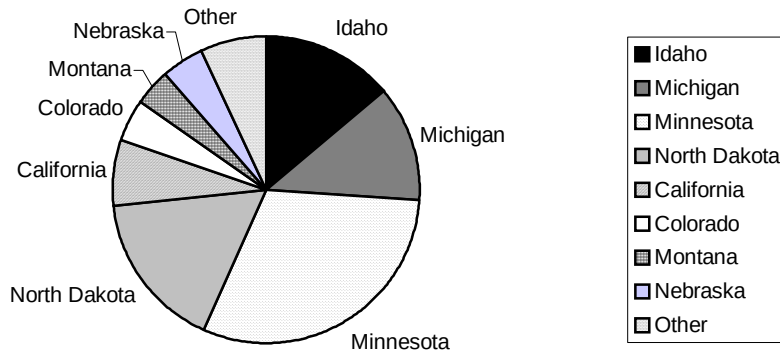
Note: Data are fiscal years. A short ton is equal to 2,000 pounds.

Source: Sugar and Sweetener Situation and Outlook. Economic Research Service, USDA (September 1999).

Exhibit 4: Sugarcane Harvests by State (%)

Note: Percentages of sugar cane acres harvested by state, 1999 crop year.

Source: Sugar and Sweetener Situation and Outlook. Economic Research Service, USDA (September 1999).

Exhibit 5: Sugar Beet Harvests by State (%)

Note: Percentages of sugar cane acres harvested by state, 1999 crop year.

Source: Sugar and Sweetener Situation and Outlook. Economic Research Service, USDA (September 1999).

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